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Corporate Internship Program

Interim Report & AI Solution Design (AS2)

Operational Analysis of RFQ Workflow & Procurement Information Management at NEGBUY

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EXECUTIVE SUMMARY

This Interim Corporate Internship Program Report (AS2) documents the analytical progress and operational findings emerging from an internship conducted at Negbuy, a platform-based industrial procurement and sourcing organization, under the procurement and supply chain function. The internship is supervised by Mr. Sayeem Salmani, Supply Chain Manager, and the project is being pursued under the academic mentorship of Dr. Gunjan Sood at the Fortune Institute of International Business.

The central operational problem identified during the internship is the absence of a structured procurement information management system at Negbuy. The procurement workflow is entirely manual, with procurement personnel receiving Requests for Quotation (RFQs) in fragmented formats, including photographs, screenshots, PDFs, Excel sheets, and text messages, and manually interpreting, processing, and tracking them across two structurally distinct sourcing streams. The first stream covers platform-based commodity sourcing through Alibaba, while the second addresses specialized industrial and turbine spare part procurement from global OEM distributors.

The analytical work completed to date includes full current-state workflow mapping across both procurement streams, identification of seven primary operational inefficiencies, and application of the Fishbone Root Cause Analysis and 5 Why frameworks to trace the structural origins of these inefficiencies. The root cause analysis reveals that the central systemic failure is the absence of any organizational procurement memory infrastructure, which forces repeated manual sourcing cycles and prevents the accumulation of supplier and RFQ intelligence over time.

As the Intellectual Contribution (IC) component of the project, the study is progressing toward the conceptual design of ProcureIQ, an AI-assisted procurement information management prototype intended to interpret multi-format RFQs, extract structured procurement data, and maintain a searchable centralized database. ProcureIQ is positioned as an operational information management tool and not as an autonomous procurement decision engine.

TABLE OF CONTENTS

Executive Summary ii
List of Tables iii
List of Figures iv
Chapter 1, Organizational and Internship Overview 1
1.1 Organization Overview, Negbuy 1
1.2 Internship Role and Functional Assignment 2
1.3 Complete RFQ Workflow at Negbuy 2
1.4 Two Procurement Streams 4
Chapter 2, Problem Identification and Operational Analysis 5
2.1 Existing Procurement Inefficiencies 5
2.2 Root Cause Analysis 7
2.3 Process Mapping of the RFQ Workflow 9
2.4 Procurement Information Flow Analysis 10
2.5 Operational Impact Assessment 11
Chapter 3, Framework-Based Operational Analysis 13
3.1 Process Mapping Framework 13
3.2 Fishbone Root Cause Analysis 14
3.3 5 Why Analysis 15
3.4 Bottleneck Analysis 16
3.5 Operational Gap Analysis 17
3.6 Information Flow Analysis 18
Chapter 4, ProcureIQ: AI-Assisted Procurement Information Management 19
4.1 Introduction to ProcureIQ 19
4.2 Problem the System Intends to Address 20
4.3 Proposed System Workflow 20
4.4 Supported RFQ Input Formats 21
4.5 Proposed Technical Architecture 22
4.6 Prototype Development Progress 23
4.7 Future Scope 23
Chapter 5, Interim Findings, Progress, and IC Calibration 24
5.1 Work Completed to Date 24
5.2 Key Operational Findings 25
5.3 Changes from Initial Proposal 25
5.4 Current IC Progress 26
5.5 Pending Work and Next Milestones 26
Chapter 6, Risks, Constraints, and Ethical Considerations 29
Chapter 7, Conclusion 31
References 32
Annexure A, Interim Evidence Log 34
Annexure B, RFQ Format Classification Matrix 35

List of Tables

Table 1: Negbuy Organizational Profile 5
Table 2: RFQ Procurement Streams, Comparative Summary 9
Table 3: Procurement Inefficiency Register 10
Table 4: 5 Why Analysis, Procurement Memory Gap 13
Table 5: Operational Gap Analysis Matrix 19
Table 6: ProcureIQ Database Schema 24
Table 7: Interim Progress Summary 26
Table 8: IC Milestone Tracker 28
Table 9: Risk and Constraint Register 29
Table 10: Interim Evidence Log 33

List of Figures

Figure 1: Current-State RFQ Workflow, Process Flow Diagram 8
Figure 2: Fishbone Root Cause Analysis, Procurement Information Fragmentation 12
Figure 3: ProcureIQ System Architecture Diagram 23

CHAPTER 1: ORGANIZATIONAL AND INTERNSHIP OVERVIEW

1.1 Organization Overview, Negbuy

Negbuy is a platform-based procurement and sourcing organization engaged in the identification, sourcing, and supply of industrial spare parts, maintenance, repair, and operations (MRO) components, industrial automation products, and specialized engineering assemblies. Operating in a demand-driven sourcing model, Negbuy functions as a procurement intermediary: clients submit requirements in the form of Requests for Quotation (RFQs), and the procurement team is responsible for identifying suitable suppliers, obtaining competitive quotations, and coordinating the fulfilment process. The organization does not maintain a physical inventory but operates on a make-to-order sourcing logic in which each RFQ triggers a fresh procurement cycle.

The procurement and supply chain function at Negbuy operates at the intersection of industrial knowledge, supplier network management, and procurement information handling. The team regularly engages with international suppliers across China, Europe, the United States, and Australia, depending on the nature of the RFQ and the availability of the required component. The procurement environment is characterized by high technical complexity, format heterogeneity in incoming RFQs, and significant reliance on individual operator expertise for product identification and supplier search decision-making.

Table 1: Negbuy Organizational Profile

Organization Name	Negbuy
Business Model	Platform-based procurement and sourcing intermediary
Functional Domain	Industrial spare parts, MRO, automation components, engineering assemblies
Sourcing Geography	China (Alibaba), Europe, USA, Australia (OEM/distributor)
Client Engagement	RFQ-based, demand-driven sourcing
Inventory Model	Non-inventory; make-to-order procurement
Procurement Team	Supply chain manager, procurement executives, interns
Key Operational Challenge	Manual, fragmented RFQ handling with no centralized information system

1.2 Internship Role and Functional Assignment

The student is placed at Negbuy as a Procurement and Purchase Intern within the procurement and supply chain function, reporting directly to Mr. Sayeem Salmani, Supply Chain Manager. The internship commenced in May 2026 and spans a period sufficient to engage meaningfully with live procurement operations across both sourcing streams. The functional assignment encompasses the full range of operational procurement activities, including RFQ receipt and interpretation, product identification, supplier identification and outreach, quotation collection and comparison, procurement coordination, and supplier follow-up management.

The nature of the internship is operationally immersive: the student is directly involved in active procurement cycles rather than observing from the periphery. This immersion has enabled detailed first-hand observation of the workflow, its bottlenecks, and the information management challenges that constitute the core of the present project. Each working day involves the receipt of new RFQs, engagement with suppliers across multiple time zones, management of follow-up communication, and coordination with the sales team regarding quotation outcomes. This operational depth provides the evidentiary foundation for the analytical work presented in this report.

1.3 Complete RFQ Workflow at Negbuy

The RFQ process at Negbuy follows a sequence that is operationally consistent with the general procurement cycle described in supply chain management literature, wherein procurement activities begin with requirement identification and proceed through supplier selection, quotation, and order confirmation (Lysons & Farrington, 2016). At Negbuy, however, this cycle is executed entirely through manual processes, and the workflow has evolved organically around individual operator capability rather than through a designed information system.

The workflow proceeds through the following stages. In the first stage, a client submits a procurement requirement to the sales team at Negbuy, either through email communication, WhatsApp, or direct contact. The requirement may be expressed as a formal RFQ document, a list of part numbers, a photograph of a product label, a screenshot of a catalogue page, or a verbal description converted into a text message. The format of the incoming client requirement is therefore highly variable and is determined entirely by the client's own communication preference.

In the second stage, the sales team reviews the client requirement and forwards it to the procurement team. At this point, the requirement is typically transmitted through an internal WhatsApp group or email, and it arrives in whatever format the client originally used. No standardization of the incoming RFQ occurs at this stage: the procurement team receives the information in the same heterogeneous format in which it was submitted by the client.

In the third stage, the procurement intern or executive receives the forwarded RFQ and manually interprets its content. This interpretation involves reading the document, photograph, or text and identifying the key procurement parameters: the manufacturer, the part number or reference, the technical specifications, the quantity required, and any equipment compatibility references. For straightforward industrial components, this stage may take a few minutes. For specialized turbine or engineering assembly components with extensive OEM references and multi-drawing linkages, interpretation can occupy a significant portion of a working day.

In the fourth stage, the intern identifies the appropriate sourcing route based on the interpreted product category and technical complexity. Standard industrial components are typically searched on Alibaba, while specialized or OEM components require Google-based distributor searches, direct manufacturer website queries, or engagement with known industrial distribution networks. ChatGPT has been used informally as a supplementary identification tool for ambiguous or partially legible RFQs.

In the fifth stage, the intern contacts multiple suppliers through email, providing the RFQ details and requesting a quotation. The supplier outreach email is composed manually for each RFQ. Following the initial email, a follow-up is sent after approximately one to two days if no response is received, and a second follow-up after a further two to three days. This follow-up cycle is managed entirely through the intern's own email account and memory of the communication timeline.

In the sixth stage, received quotations are compared manually. The intern reviews the quotations from multiple suppliers and identifies the most competitive offer based on price, delivery timeline, and supplier credibility. The preferred quotation is then forwarded to the sales team, which marks it up and presents it to the client. If the client confirms acceptance, the procurement team verifies the order and initiates the fulfilment process.

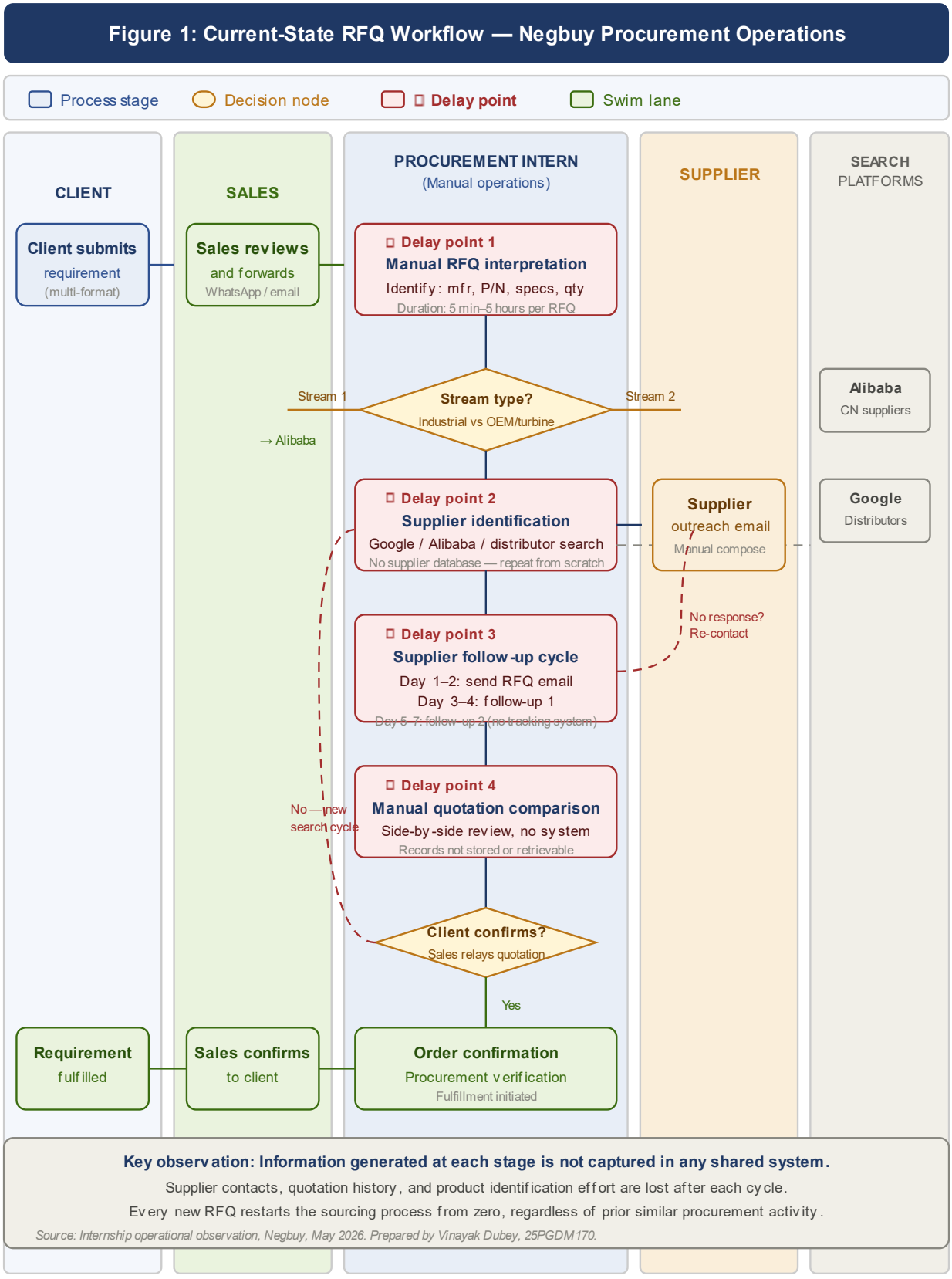


Figure 1: Current-State RFQ Workflow, Process Flow Diagram (Refer Annexure B)

1.4 Two Procurement Streams at Negbuy

A critical structural feature of the procurement environment at Negbuy is the simultaneous operation of two fundamentally distinct sourcing streams, each characterized by different RFQ formats, different sourcing methodologies, different supplier ecosystems, and different execution timelines. Understanding this dual-stream structure is essential to appreciating the complexity of the manual information management challenge.

Table 2: RFQ Procurement Streams, Comparative Summary

Dimension	Stream 1: Platform-Based Industrial Sourcing	Stream 2: Specialized OEM / Turbine Sourcing
Typical Products	PLC modules, sensors, relays, automation components, bearings, instrumentation parts	GE gas turbine spare parts, gearbox assemblies, OEM repair kits, generator components, engineering seals
RFQ Format	Part number, model number, product photograph	Technical spec sheet, OEM drawing references, multi-part assembly descriptions
Primary Sourcing Platform	Alibaba, Chinese supplier portals	Google, distributor websites, OEM portals, email outreach
Geographic Supplier Base	China (primarily)	Europe, USA, Australia
Sourcing Complexity	Moderate; structured part numbers ease identification	High; multiple OEM references, drawing numbers, and compatibility codes
Typical Cycle Time	Two to five days	Five to fifteen days or longer
Follow-Up Requirement	Limited	Extensive; multiple follow-up rounds required

The first stream, platform-based commodity and industrial procurement, involves components such as Siemens PLC modules, ABB I/O units, Yokogawa sensors, Schmersal safety relays, and similar industrial automation products. These components typically carry structured manufacturer part numbers that can be entered directly into Alibaba's search interface, yielding a set of potential suppliers whose credibility can be assessed through platform ratings and past transaction records. The sourcing cycle for this stream is relatively fast, and supplier communication tends to be efficient due to the volume-oriented nature of the Chinese export market.

The second stream, specialized industrial and turbine spare parts procurement, presents a substantially more complex operational challenge. RFQs in this category routinely reference GE gas turbine components with

equipment model numbers such as MS7001 EA, Flender gearbox assemblies with multi-code identification systems, Milton Roy pump repair kits with OEM drawing references, and LM6000PC turbine spare parts with cross-referenced manufacturer codes. These RFQs are often received as extensive tabular documents containing serial numbers, dimensional specifications, material standards, and equipment compatibility codes. Identifying the correct component, finding a legitimate distributor, and obtaining a quotation from a reputable supplier in Europe or the United States requires significant manual effort and domain knowledge.

CHAPTER 2: PROBLEM IDENTIFICATION AND OPERATIONAL ANALYSIS

2.1 Existing Procurement Inefficiencies

Inefficiency in procurement operations typically arises when information handling processes are not aligned with the volume, variety, and velocity of incoming procurement demands (Chopra & Meindl, 2016). At Negbuy, the operational context presents all three of these pressure dimensions simultaneously: RFQ volume is significant, format variety is extreme, and client expectations around response velocity are high. Against this backdrop, the manual nature of the current procurement workflow generates a compound set of identifiable inefficiencies.

Table 3: Procurement Inefficiency Register

Inefficiency	Description	Operational Impact
Manual RFQ interpretation	Each RFQ is read, decoded, and interpreted individually by the operator without tool support.	Time delay at the entry stage; error risk in identification; operator fatigue.
Repeated supplier searching	Every new RFQ triggers a fresh supplier search regardless of prior sourcing for similar components.	Significant wasted effort; loss of accumulated supplier knowledge.
No centralized procurement database	Procurement records are stored in personal email inboxes, spreadsheets, and WhatsApp threads.	No retrievability; supplier history is inaccessible across personnel.
Fragmented communication channels	RFQs arrive via WhatsApp, email, and direct message; outreach is managed across multiple accounts.	Coordination failures; missed follow-ups; information silos.
Procurement memory dependency	Sourcing knowledge resides in individual operators; no system-level retention occurs.	Knowledge loss when personnel change; repeated learning cycles.
Manual quotation comparison	Received quotations are compared by reading and cross-referencing in isolation.	Inconsistent comparison logic; time-intensive; risk of suboptimal selection.

Absent RFQ retrieval mechanism	There is no way to search whether a similar RFQ has been handled previously.	Inability to leverage historical procurement intelligence; repeated effort.

The most operationally consequential among these inefficiencies is the absence of a centralized procurement database. The lack of any structured repository means that every RFQ encountered by the procurement team is effectively processed as a novel problem, regardless of whether an identical or highly similar component has been sourced in a prior period. A supplier identified, contacted, and successfully engaged for a Siemens PLC module six weeks earlier is not recalled or referenced when a similar component arrives in a subsequent RFQ: the sourcing cycle restarts from zero. This pattern of information loss and repeated effort constitutes what may be termed the procurement memory gap, an organizational condition in which procurement intelligence is generated continuously but retained at the individual rather than the institutional level.

The fragmentation of communication channels compounds this problem significantly. Because RFQs arrive through multiple informal channels, WhatsApp messages, forwarded email attachments, PDF documents dropped into shared folders, and because outreach to suppliers is conducted through individual email accounts, the organizational procurement footprint is invisible to anyone other than the specific individual who handled a given RFQ. There is no shared visibility into which suppliers have been contacted, which have responded, what quotation prices have been received, or what the outcome of a given sourcing cycle was. This absence of shared procurement intelligence is a structural driver of inefficiency that no amount of individual effort can compensate for.

2.2 Root Cause Analysis

2.2.1 Fishbone (Ishikawa) Analysis

The fishbone diagram, or Ishikawa cause-and-effect diagram, is a widely applied quality management tool used to systematically trace the root causes of an operational problem across multiple causal dimensions (Slack et al., 2019). Applied to the procurement information management problem at Negbuy, the fishbone analysis identifies causal contributors across the dimensions of people, processes, technology, information, and organizational design.

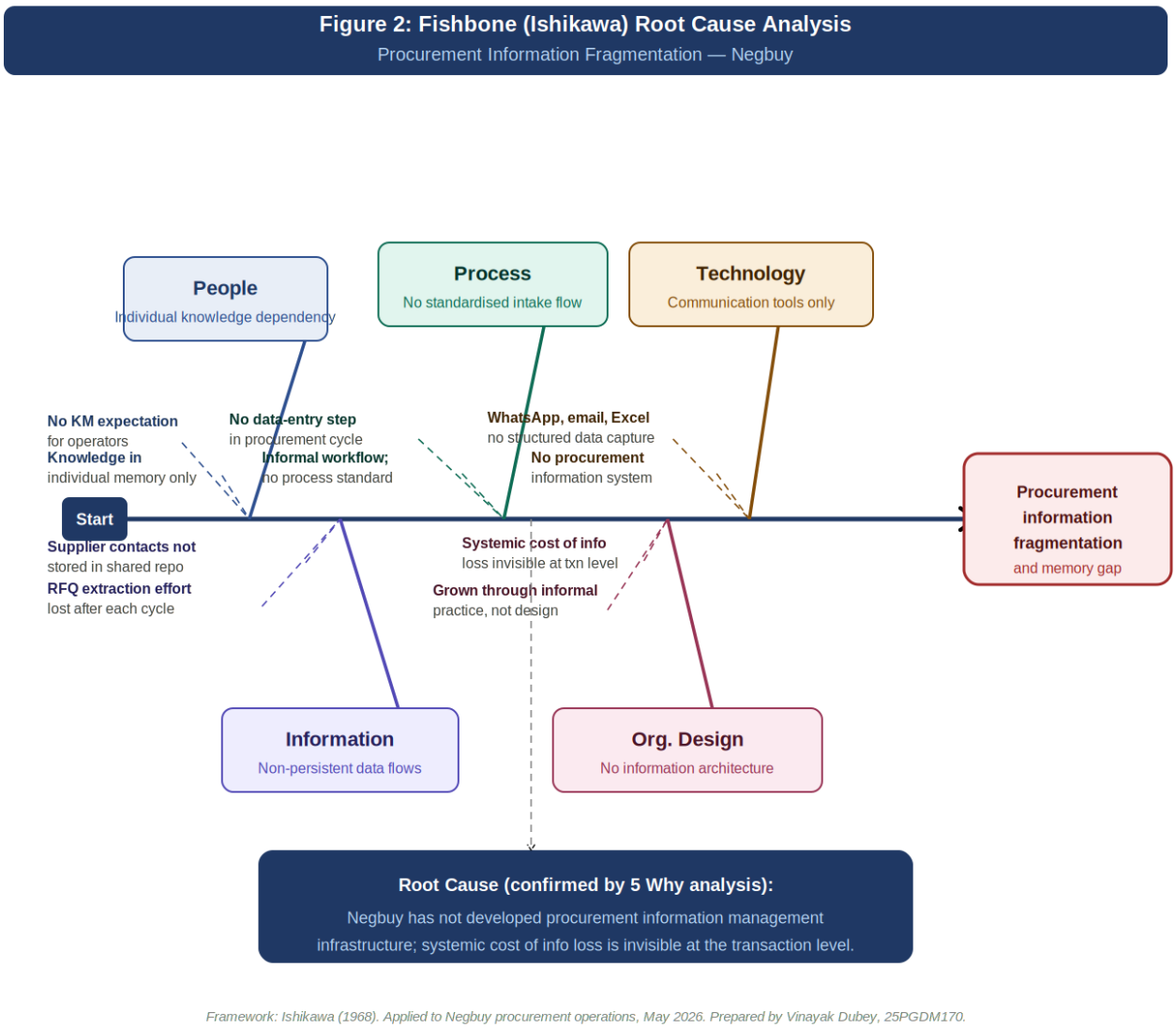


Figure 2: Fishbone Root Cause Analysis, Procurement Information Fragmentation (Refer Annexure B)

In the people dimension, the root cause lies in the absence of any formal knowledge management expectation for procurement operators. Individual operators are responsible for their own RFQ-to-quotation lifecycle, and there is no organizational norm or process requiring the documentation of sourcing activities in a retrievable format. The consequence is that procurement knowledge accumulates in individual memory and personal file storage rather than in organizational systems.

In the process dimension, the absence of a standardized RFQ intake process means that incoming requirements are handled in whatever format they arrive, without conversion to a common structure. There is no defined step in the current workflow for data entry into a shared system, because no such system exists. The procurement cycle, from RFQ receipt to quotation submission, is executed entirely through informal channels and personal judgement.

In the technology dimension, the organization currently relies on WhatsApp, email, Alibaba's supplier portal, Google search, and personal Excel files as the totality of its procurement technology stack. None of these tools provides the structured data capture, retrieval, or organizational memory functions that a procurement information system would offer. The tools in use were designed for communication and search rather than procurement intelligence management.

In the information dimension, the root cause is the non-standardized and non-persistent nature of procurement data. Information extracted from RFQs, manufacturer names, part numbers, technical specifications, is used once during the sourcing cycle and then effectively discarded. There is no process by which this extracted information is stored in a form accessible to future procurement cycles.

2.2.2 Five Why Analysis

The 5 Why technique, originally developed within the Toyota Production System, is an iterative interrogative method designed to identify the fundamental cause of a problem by repeatedly asking why a condition exists (Ohno, 1988). Applied to the central problem of repeated manual sourcing cycles at Negbuy, the analysis yields the following causal chain.

Table 4: 5 Why Analysis, Procurement Memory Gap at Negbuy

Level	Question	Answer
Problem	Why does the procurement team repeat supplier searches for similar RFQs?	Because there is no record of previous sourcing activities for reference.
Why 1	Why is there no record of previous sourcing activities?	Because procurement records are not stored in any shared or retrievable system.
Why 2	Why are procurement records not stored in a shared system?	Because no structured procurement database or information management system exists at Negbuy.
Why 3	Why does no such system exist?	Because procurement activities have grown through informal practice rather than designed information architecture.
Why 4	Why has no information architecture been designed?	Because the manual workflow has been operationally sufficient for individual cycles, masking the systemic cost of repeated effort.
Root Cause	Underlying finding	The organization has not developed a procurement information management system because the cumulative cost of information loss is not visible at the individual transaction level.

The 5 Why analysis reveals that the root cause is not a failure of individual effort but a structural absence: Negbuy has not yet developed the organizational infrastructure to retain and retrieve procurement intelligence. This finding is critical because it shifts the framing of the problem from a personnel capability issue to an information systems design challenge, a distinction that is highly relevant to the proposed ProcureIQ intervention.

2.3 Process Mapping of the RFQ Workflow

Process mapping is a foundational operations management tool used to represent the sequence of activities, decision points, and information flows that constitute a business process (Harrington, 1991). A current-state process map of the RFQ workflow at Negbuy reveals seven discrete process stages, three significant decision nodes, and four identifiable delay points.

The process map identifies the following primary delay points: the RFQ interpretation stage, where manual identification of technical specifications can occupy considerable time for complex components; the supplier identification stage, where the absence of a supplier database requires fresh searches for each RFQ; the supplier response stage, where international suppliers may take several days to respond, requiring multiple follow-up cycles; and the quotation comparison stage, where manual side-by-side review of received quotations adds time and cognitive load to the process.

Across both procurement streams, the cumulative delay introduced by these manual stages is significant. For Stream 2 (specialized industrial and turbine components), the time from RFQ receipt to quotation submission can extend to several weeks, during which the procurement team's attention is divided across multiple concurrent RFQs in various stages of the sourcing cycle. The absence of a tracking system means that the management of concurrent RFQs relies entirely on individual memory and informal reminders, creating a high risk of missed follow-ups and delayed responses.

2.4 Procurement Information Flow Analysis

Information flow analysis examines how data moves through an operational system, identifying points at which information is created, transformed, transferred, and, critically, lost (Davenport & Prusak, 1998). At Negbuy, the procurement information flow is characterized by a sequence of creation and loss events that collectively produce the procurement memory gap identified in the root cause analysis.

Information is created at the point of RFQ receipt, when the client submits a procurement requirement containing potentially valuable product identification data. This information is then partially extracted during the manual

interpretation stage, where the operator identifies the manufacturer and part number but typically records only what is immediately needed for the sourcing search. The remainder of the information, drawing references, equipment compatibility codes, alternative part numbers, technical dimensions, may be referenced during the sourcing cycle but is not systematically captured.

Information is further created during the supplier outreach phase, when the operator identifies viable suppliers and records their contact details. However, this information is stored in the operator's email account and is not transferred to any shared repository. When a supplier that was successfully engaged for one RFQ becomes relevant for a subsequent RFQ handled by a different operator, the prior contact record is inaccessible. The supplier identification effort is therefore duplicated.

Information is created again when quotations are received, but quotation comparison and selection are performed manually without structured recording of the comparison logic or the rejected alternatives. Over time, this means that the organization has no basis for analysing supplier pricing patterns, lead time performance, or response reliability, all of which would be valuable inputs for future procurement strategy.

2.5 Operational Impact Assessment

The cumulative operational impact of the inefficiencies identified above can be assessed across three dimensions: time cost, knowledge cost, and strategic cost. In the time dimension, the manual nature of each stage of the procurement cycle, from interpretation through supplier identification, outreach, follow-up, and quotation comparison, imposes a significant per-RFQ time burden on procurement staff. While precise time measurements are beyond the scope of the current study, observational evidence from the internship suggests that a complex Stream 2 RFQ can occupy several hours of active work spread across multiple days, with a substantial proportion of that effort devoted to activities that would be unnecessary if a structured procurement information system existed.

In the knowledge dimension, the operational impact is the persistent erosion of procurement intelligence. Each completed sourcing cycle generates knowledge, about the product, the supplier, the quotation, and the sourcing pathway, that is immediately at risk of being lost when the operator moves to the next task. This erosion means that Negbuy's effective procurement capability does not improve over time in proportion to the volume of RFQs handled: the organization is perpetually at an early stage of learning for any product category that has not been recently sourced.

In the strategic dimension, the absence of supplier relationship continuity represents a missed opportunity for procurement advantage. Research in supply chain management consistently demonstrates that sustained supplier relationships generate cost reductions, lead time improvements, and preferential access to constrained supply (Cousins et al., 2008). At Negbuy, supplier relationships are effectively reset with each new sourcing cycle, preventing the accumulation of relational capital that would otherwise improve procurement performance.

CHAPTER 3: FRAMEWORK-BASED OPERATIONAL ANALYSIS

3.1 Process Mapping Framework

Process mapping provides operations managers with a structured method for visualising the activities, decision points, and information flows that constitute a business process, thereby creating the analytical foundation for identifying redundancies, delays, and improvement opportunities (Slack et al., 2019). Applied to the Negbuy procurement workflow, the process mapping exercise documented in Chapter 1 yields several analytically significant observations.

The process map reveals that the Negbuy RFQ workflow is operationally linear in its basic structure, client requirement, procurement team processing, supplier outreach, quotation collection, and sales coordination, but that the execution of each stage is characterized by significant variability that is driven by format heterogeneity and the absence of decision support tools. The interpretation stage, in particular, has no defined duration or structured output format: an operator may spend five minutes or five hours on a single RFQ depending on the complexity of the component and the legibility of the incoming document.

Process mapping also reveals that the current workflow contains no feedback loop: information generated during one procurement cycle does not re-enter the process at the start of the next. This absence of feedback is analytically significant because it confirms that the process is designed, implicitly, through the absence of formal design, as a set of independent, unconnected transactions rather than as a learning system. The managerial implication is that workflow improvement alone, without the addition of an information retention mechanism, will not address the fundamental inefficiency.

3.2 Fishbone Root Cause Analysis Application

The Fishbone analysis described in Section 2.2.1 is applied here at the framework level to consolidate the causal picture of procurement information fragmentation at Negbuy. The five causal dimensions identified through the analysis, people, process, technology, information, and organizational design, collectively confirm that the procurement information problem is systemic rather than symptomatic. It cannot be resolved through training, individual effort, or incremental process improvement alone; it requires a structural intervention in the form of an information management system.

The framework application also reveals that the technology dimension is the most tractable entry point for intervention. While people, process, and organizational design dimensions involve cultural and structural change that requires longer timelines and broader organizational commitment, the technology dimension can be

addressed through the design and deployment of a purpose-built procurement information management tool. This observation directly motivates the ProcureIQ concept developed in Chapter 4.

3.3 Five Why Analysis Application

The 5 Why analysis presented in Table 4 demonstrates that the root cause of the procurement memory gap is not operator negligence or insufficient training but rather the absence of an organizational norm and infrastructure for procurement knowledge retention. This finding has important implications for the proposed intervention: any solution that places the burden of data entry and knowledge retention on individual operators without reducing the cognitive load of that task will encounter the same adoption barriers that have prevented informal knowledge-sharing efforts from gaining traction in the current environment.

The ProcureIQ system addresses this root cause directly by automating the data extraction step, the step at which procurement knowledge is most commonly lost, and eliminating the need for manual data entry by the operator. If the system can read an RFQ and populate a structured database record automatically, the knowledge retention process requires no additional operator effort beyond the RFQ processing activity that already takes place.

3.4 Bottleneck Analysis

Bottleneck analysis, rooted in the Theory of Constraints developed by Goldratt, identifies the stage in a process that limits the overall throughput of the system and therefore constrains operational performance (Goldratt & Cox, 2004). In the Negbuy procurement workflow, the primary bottleneck is located at the RFQ interpretation and supplier identification stage, not at the supplier outreach or quotation collection stages. This is a significant analytical finding.

The reason the interpretation and identification stage constitute the primary bottleneck is that it is entirely dependent on individual operator knowledge and cannot be parallelized or delegated without replicating the manual effort. When the procurement team receives multiple RFQs simultaneously, as frequently occurs, the operator must process them sequentially through the interpretation stage, creating a queue that delays the initiation of supplier outreach for all but the first RFQ in the sequence. Improving the speed and consistency of the interpretation stage through AI-assisted extraction would therefore increase the throughput of the entire procurement process rather than merely improving one isolated activity.

3.5 Operational Gap Analysis

Table 5: Operational Gap Analysis Matrix

Operational Dimension	Current State	Desired State	Gap
RFQ Interpretation	Manual, format-dependent, operator-variable	Structured, automated extraction from any format	No AI extraction tool; no standard output format
Procurement Data Storage	Personal email, Excel files, WhatsApp	Centralized, searchable, structured database	No centralized system exists
Supplier Intelligence	Resides in individual memory	Shared, retrievable, historically accumulated	No shared supplier repository
RFQ Retrieval	Not possible	Full-text and field-level search of historical RFQs	No retrieval mechanism exists
Quotation Tracking	Manual, ad hoc	Structured, date-stamped, comparative	No systematic quotation record
Process Consistency	Variable, operator-dependent	Standardized across operators and streams	No process standard or template

The operational gap analysis confirms that the current and desired states diverge across all six operational dimensions examined. Critically, the gap is widest in the areas of procurement data storage and RFQ retrieval, the two dimensions most directly addressable by the ProcureIQ system. The analysis also indicates that the desired state does not require advanced artificial intelligence in the predictive or autonomous sense; it requires structured data capture, reliable storage, and efficient retrieval, capabilities that are achievable with current technology and appropriate system design.

3.6 Information Flow Analysis

Information flow analysis, grounded in knowledge management theory, examines how organizational knowledge is created, encoded, stored, and retrieved across operational contexts (Nonaka & Takeuchi, 1995). At Negbuy, the information flow analysis conducted during the internship reveals a pattern in which procurement knowledge is created repeatedly but encoded rarely, resulting in a persistent gap between the organization's cumulative procurement experience and its accessible procurement intelligence.

The analysis identifies three critical information flow failures. The first is the non-encoding of RFQ interpretation outcomes: when an operator successfully identifies a complex turbine component by decoding a multi-reference

technical description, that identification effort is not recorded in any retrievable form. The second is the non-persistence of supplier contact records: supplier email addresses, contact names, and response histories identified during outreach are not entered into any shared repository. The third is the non-aggregation of quotation data: received quotations are used for the immediate sourcing decision but are not accumulated into a database that would enable price benchmarking or supplier performance assessment.

These three information flow failures collectively constitute the procurement intelligence deficit that the ProcureIQ concept is designed to address.

CHAPTER 4: PROCUREIQ, AI-ASSISTED PROCUREMENT INFORMATION MANAGEMENT

4.1 Introduction to ProcureIQ

ProcureIQ is proposed as a lightweight AI-assisted procurement information management system designed to address the operational inefficiencies identified in the preceding chapters. The system is not conceived as an autonomous procurement engine, a predictive sourcing AI, or a supplier negotiation tool. It is conceived as an information structuring and retention tool: a system that reads incoming RFQs from whatever format they arrive in, extracts structured procurement data, organizes that data into a centralized searchable repository, and makes accumulated procurement intelligence available for retrieval in future sourcing cycles.

The conceptual positioning of ProcureIQ aligns with the broader literature on AI-assisted information management in operations, which distinguishes between AI systems that make decisions and AI systems that structure information to support human decision-making (Davenport & Ronanki, 2018). ProcureIQ is firmly in the latter category: it supports the procurement team by eliminating the manual effort of data extraction and storage, without attempting to automate the sourcing judgement that remains appropriately with the human operator.

4.2 Problem the System Intends to Address

ProcureIQ directly addresses the procurement memory gap identified through the root cause analysis. The system is designed to ensure that the procurement intelligence generated during each sourcing cycle, including the product identification, supplier identification, contact records, and quotation data, is captured, structured, and retained in a form accessible to future procurement activities. By eliminating the information loss that currently occurs at the conclusion of each sourcing cycle, ProcureIQ aims to transform the procurement function from a collection of independent transactions into a learning system that improves its performance as its database grows.

The system also addresses the format fragmentation challenge by accepting RFQ inputs in all formats currently encountered in the Negbuy workflow. Rather than requiring clients or the sales team to submit RFQs in a standardized format, a requirement that would face significant adoption resistance, ProcureIQ accepts images, PDFs, Excel files, screenshots, and text descriptions and handles the normalization internally through AI-assisted extraction.

4.3 Proposed System Workflow

The ProcureIQ workflow is designed in five sequential stages that mirror the existing manual procurement cycle while automating the information-intensive components:

In Stage 1, the user uploads an RFQ to the ProcureIQ interface. The input can be a photograph, a PDF document, an Excel file, a screenshot, or plain text. The system accepts the input without requiring pre-processing or format conversion by the user.

In Stage 2, the AI extraction engine processes the uploaded input and identifies the key procurement parameters: the manufacturer name, part number or reference, product category, quantity, technical specifications, equipment compatibility references, and any OEM drawing or serial number references present in the document.

In Stage 3, the extracted information is displayed to the user in a structured table for review and confirmation. The user can correct any extraction errors before the data is committed to the database, ensuring that the stored records are accurate.

In Stage 4, the confirmed data is automatically written to the centralized procurement database, currently envisaged as a structured Google Sheet with defined field columns. Each new RFQ adds one or more rows to the database, building the procurement intelligence repository incrementally with each cycle.

In Stage 5, the user can query the database using a search interface, entering a manufacturer name, part number, or product keyword to retrieve historical records. The retrieval function shows all prior RFQs matching the search criteria, including the suppliers previously contacted, the quotations received, and the status of those procurement cycles.

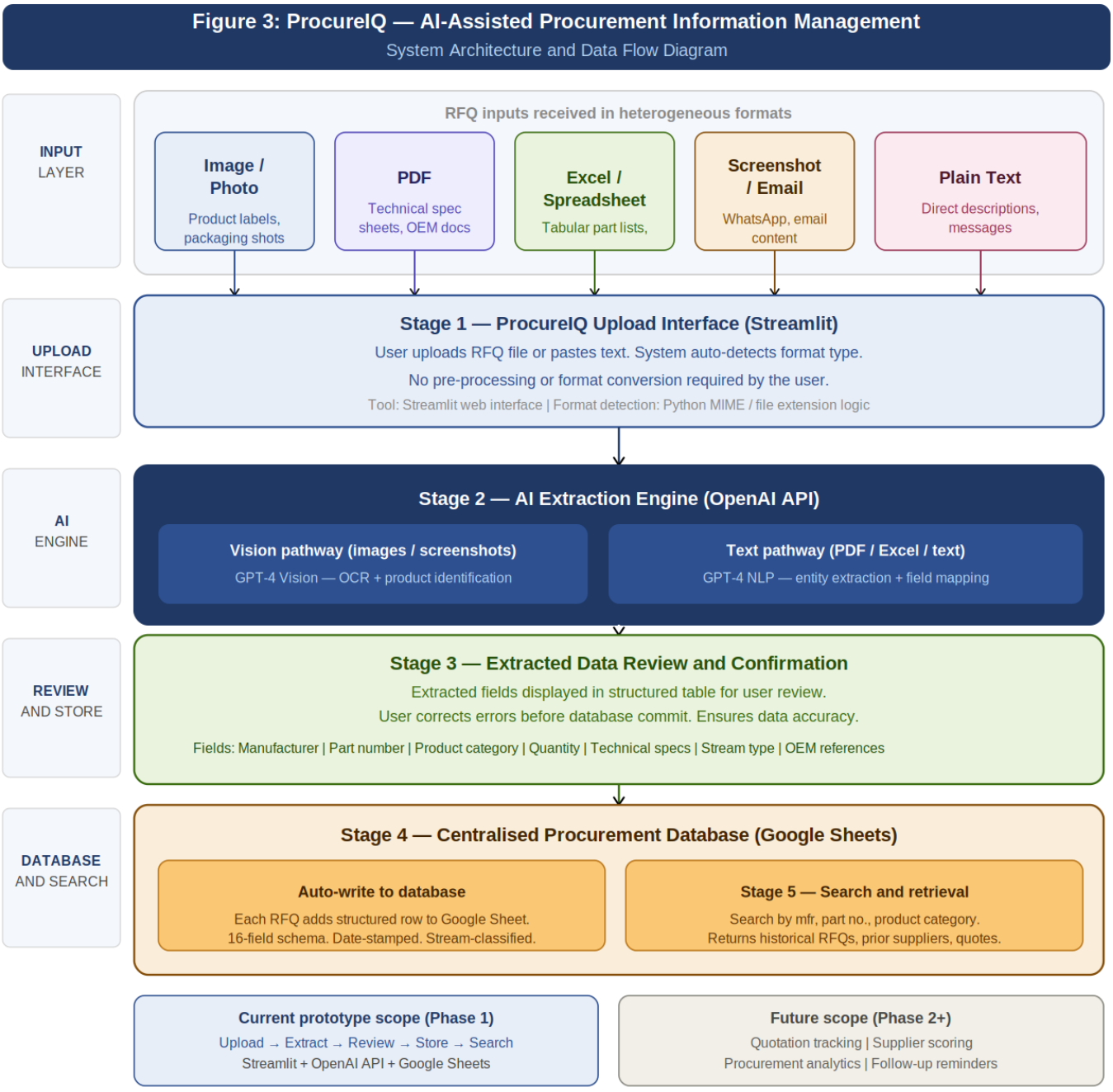


Figure 3: ProcureIQ System Architecture Diagram (Refer Annexure B)

4.4 Supported RFQ Input Formats

The ProcureIQ system is designed to accept the full range of RFQ formats currently encountered at Negbuy, ensuring that no change to the upstream client or sales team workflow is required for the system to deliver value. The supported input formats in the initial scope of the prototype are:

- Photographs and images of product labels, packaging, and components
- PDF documents containing technical specifications, OEM references, and part lists
- Excel files with tabular RFQ data including part numbers and quantities

- Screenshots of WhatsApp messages, email content, and catalogue pages
- Plain text descriptions of required components submitted via messaging platforms

The ability to process image-based inputs is particularly important in the Negbuy context because a significant proportion of Stream 1 RFQs arrive as photographs of product labels, packaging, or component close-ups. These images contain the manufacturer and part number in a format that is visually legible but not digitally searchable. The AI extraction engine must be capable of performing optical character recognition on these images and mapping the extracted text to the structured database fields.

4.5 Proposed Technical Architecture

Table 6: ProcureIQ Database Schema, Core Fields

Field Name	Data Type	Description	Example Value
RFQ_ID	Auto-generated	Unique identifier for each RFQ entry	RFQ-2026-001
Date_Received	Date	Date the RFQ was received	15 May 2026
Input_Format	Text	Format of the original RFQ input	Image / PDF / Excel / Text
Manufacturer	Text	Identified manufacturer name	Siemens / GE / ABB
Part_Number	Text	Manufacturer part or reference number	6ES7307-1EA01-0AA0
Product_Category	Text	Functional category of the component	PLC Power Supply / Turbine Seal
Technical_Specs	Text	Key technical specifications extracted	Dimensions, material, compatibility
Quantity	Number	Required quantity	20 EA
Stream	Text	Procurement stream classification	Stream 1 (Alibaba) / Stream 2 (OEM)
Supplier_Contacted	Text	Name of supplier contacted	[Supplier Name]
Supplier_Email	Text	Email address of supplier	[Anonymized]
Contact_Date	Date	Date supplier was first contacted	16 May 2026
Quotation_Received	Yes/No	Whether a quotation was received	Yes
Quotation_Amount	Currency	Quoted price (anonymized in academic outputs)	[Amount]
Status	Text	Current status of the RFQ	Pending / Quoted / Confirmed

Notes	Text	Additional observations or follow-up notes	Follow-up sent 18 May

The technical architecture of the ProcureIQ prototype is built around four components: a user interface layer, an AI extraction engine, a database storage layer, and a search and retrieval module. In the prototype phase, the user interface is envisaged as a lightweight web form or Streamlit-based application that accepts file uploads and displays extracted data for confirmation. The AI extraction engine leverages the OpenAI API, specifically the GPT-4 Vision capability for image-based RFQs and GPT-4 text completion for text-based inputs, to perform the extraction and structuring task. The database layer is implemented using Google Sheets, providing a shared, accessible, and zero-infrastructure repository that can be accessed by any team member with appropriate permissions. The search and retrieval module uses Google Sheets filtering and query functions to enable keyword-based retrieval of historical records.

4.6 Prototype Development Progress

At the time of this interim report, the ProcureIQ project has progressed through the conceptual design and system architecture phases. The workflow has been fully specified, the database schema has been designed as presented in Table 6, the input format requirements have been confirmed, and the AI extraction logic has been outlined at the prompt engineering level. The technical architecture selection has been finalized, with the OpenAI API and Google Sheets combination confirmed as the implementation pathway.

Prompt engineering for the extraction task has been explored manually using live RFQ samples encountered during the internship. Initial testing confirms that GPT-4 can successfully extract manufacturer names, part numbers, and product categories from text-based RFQs with high accuracy. Image-based extraction testing is in progress. The prototype development phase, involving the construction of the Streamlit interface, the API integration, and the Google Sheets connection, is planned to begin in the subsequent weeks of the internship.

4.7 Future Scope

Several capabilities are deliberately excluded from the initial prototype scope and positioned as future scope recommendations for subsequent development phases. Quotation tracking and comparison functionality, the automated recording and comparison of supplier quotation data, is acknowledged as a high-value extension but is excluded from the current prototype to maintain scope discipline. Supplier scoring and relationship management features, which would aggregate supplier performance data across multiple RFQs to generate reliability ratings, represent a meaningful future capability but require a greater volume of historical data to be analytically useful. Procurement analytics dashboards providing management-level visibility into RFQ volume, sourcing cycle times, and supplier response rates are envisaged as the final layer of the system but are beyond the scope of the internship prototype.

CHAPTER 5: INTERIM FINDINGS, PROGRESS, AND IC CALIBRATION

5.1 Work Completed to Date

Table 7: Interim Progress Summary

Activity	Status	Remarks
Organizational and workflow immersion	Complete	Full end-to-end RFQ cycle observed across both procurement streams.
Current-state workflow mapping	Complete	Seven-stage process map documented with decision nodes and delay points.
Procurement inefficiency identification	Complete	Seven primary inefficiencies identified and recorded.
Fishbone root cause analysis	Complete	Five causal dimensions mapped; technology dimension identified as primary intervention point.
5 Why analysis	Complete	Root cause confirmed as absence of procurement information infrastructure.
Bottleneck analysis	Complete	RFQ interpretation and supplier identification confirmed as primary bottleneck.
Operational gap analysis	Complete	Six-dimension gap matrix developed.
RFQ classification framework	Complete	Stream 1 and Stream 2 classification matrix finalized.
Database schema design	Complete	Sixteen-field procurement database schema designed.
ProcureIQ system architecture	Complete	Technical architecture confirmed; implementation pathway selected.
Prompt engineering exploration	In Progress	Manual extraction testing using live RFQ samples; image extraction in progress.
Prototype development	Planned	To begin in subsequent internship phase.
AS2 Interim Report	Complete	Current document.

5.2 Key Operational Findings

The internship has generated several analytically significant operational findings that extend beyond the initial problem framing in the AS1 proposal. First, the dual-stream procurement structure is more consequential than initially anticipated: the degree of difference between Stream 1 and Stream 2 in terms of RFQ complexity, sourcing methodology, and cycle time means that a single solution must be flexible enough to handle both contexts without forcing either stream into an inappropriate workflow. The ProcureIQ system design has been calibrated to accommodate this dual-stream reality.

Second, the informal use of ChatGPT by procurement operators for RFQ interpretation represents an important organizational signal: the team has independently identified AI assistance as a useful tool for the interpretation challenge, even in the absence of any formal system. This informal adoption validates the AI-assisted approach and suggests that the ProcureIQ concept will encounter a receptive user environment.

Third, the supplier follow-up cycle is a more significant operational burden than initially appreciated. The requirement to send multiple follow-up emails to international suppliers, manage response timelines across different time zones, and coordinate the follow-up cycle across concurrent RFQs without a tracking system creates a substantial and largely invisible administrative burden that is a strong candidate for future automation.

5.3 Changes from the Initial Proposal

The initial AS1 proposal accurately identified the core operational problem and the general direction of the project. However, the internship experience has led to several refinements in analytical focus and project scope. The dual-stream procurement structure was identified in the AS1 proposal but its operational significance has become more pronounced in the interim analysis, leading to greater emphasis on the stream-specific design requirements of the ProcureIQ system. The bottleneck analysis and information flow analysis frameworks, while anticipated in the AS1, have been applied more rigorously in the interim phase, producing the operational gap analysis matrix and the information flow failure taxonomy presented in Chapter 3. The database schema, only conceptually referenced in the AS1, has been fully designed in the interim phase. These refinements represent analytical deepening rather than directional change.

5.4 Current IC Progress

Table 8: IC Milestone Tracker

IC Milestone	Target Date	Status
Problem identification and operational analysis complete	End of Week 6	Complete
Framework application complete	End of Week 7	Complete
ProcureIQ conceptual design and database schema finalized	End of Week 8	Complete
Prompt engineering and extraction testing	Weeks 8–10	In Progress
Prototype development (Streamlit + API + Sheets)	Weeks 10–12	Planned
Prototype testing with live RFQ samples	Week 12–13	Planned
IC case study draft	Week 13–14	Planned
Final IC submission	Session 21	Planned

5.5 Pending Work and Next Milestones

The primary deliverable for the next phase of the internship is the development of the ProcureIQ prototype. This involves three parallel workstreams: the construction of the Streamlit-based user interface, the integration of the OpenAI API extraction logic, and the design and connection of the Google Sheets database backend. Each of these workstreams has been scoped and planned, and the technical dependencies have been identified. The next academic milestone is the AS1 revised proposal submission, which will incorporate the refined scope, updated timeline, and IC progress documentation generated during the interim phase. The AS2 Interim Report (current document) represents the primary academic deliverable of the current stage.

CHAPTER 6: RISKS, CONSTRAINTS, AND ETHICAL CONSIDERATIONS

6.1 Risk and Constraint Register

Table 9: Risk and Constraint Register

Risk / Constraint	Category	Likelihood	Impact	Mitigation Strategy
Limited access to historical procurement records	Data	High	Medium	Project relies on live RFQ data from the internship period; longitudinal analysis is explicitly excluded from scope.
API rate limits and cost constraints for OpenAI integration	Technical	Medium	Medium	Prototype will be designed for limited-volume demonstration; full-scale deployment is outside internship scope.
Supplier and client confidentiality restrictions	Ethical	High	Low	All data used in academic outputs will be anonymized; no commercially sensitive information will be reproduced.
Internship timeline constraint for prototype completion	Timeline	Medium	High	Prototype scope is deliberately limited to core extraction and storage functions; advanced features deferred to future scope.
Potential change in internship role or responsibilities	Operational	Low	High	Academic work is grounded in observations already completed; project can proceed with existing evidence if operational exposure changes.
AI extraction accuracy for complex technical RFQs	Technical	Medium	Medium	Manual review and confirmation step is built into the ProcureIQ workflow to catch and correct extraction errors.

6.2 Internship Constraints

The primary constraint affecting the project is the internship duration, which limits the time available for prototype development, testing, and refinement. The analytical work, workflow mapping, framework application, and conceptual system design, has been completed within the available time, but the prototype development phase must be managed carefully to ensure that a functional demonstration can be produced before the final submission deadline. The scope of the prototype has been calibrated accordingly, prioritizing the core extraction and storage functionality over the more complex search and retrieval features.

6.3 Data Availability Constraints

The project relies primarily on live RFQ data from the internship period. Historical procurement records from before the internship commencement date are not accessible, which means that the database schema and retrieval

functionality cannot be fully tested with a rich historical dataset. This constraint is acknowledged and does not affect the conceptual validity of the system design, but it does limit the depth of the retrieval demonstration that can be presented in the final submission.

6.4 Ethical and Confidentiality Considerations

The project raises several ethical considerations that have been proactively addressed throughout the internship. All RFQ data, supplier contact information, and quotation records used in academic analysis or prototype testing are anonymized. Manufacturer names and part numbers may be referenced in the academic report where necessary for analytical clarity, but client names, supplier contact details, and commercial pricing information are treated as confidential and are not reproduced in academic outputs.

The use of an AI-based extraction system raises the additional question of data handling responsibility, particularly with respect to the transmission of potentially sensitive procurement information to a third-party API provider (European Parliament, 2016). This concern is noted and will be addressed in the prototype design through the use of anonymized or synthetic test data for API testing, and through the implementation of appropriate data handling protocols for any live data processed during the demonstration.

The project complies fully with FIIB academic integrity requirements. All sources are cited in APA 7th Edition format, the similarity report is appended as required, and the AI content detection result is noted. The analytical content of this report is grounded in original operational observation and academic synthesis rather than reproduction of prior work.

CHAPTER 7: CONCLUSION

This interim report documents the analytical progress of a Corporate Internship Program project investigating the operational challenges of RFQ workflow management and procurement information handling at Negbuy. The internship experience has provided a rich evidentiary foundation for the analysis, grounded in direct participation in active procurement cycles across two structurally distinct sourcing streams.

The central operational finding is unambiguous: the procurement function at Negbuy operates under a persistent procurement memory gap, in which the intelligence generated during each sourcing cycle is not retained in any retrievable form, forcing the repetition of manual effort and preventing the accumulation of institutional procurement knowledge. This gap is structural rather than personal, it is a product of the absence of any information management infrastructure, and it requires a structural rather than behavioral intervention.

The framework-based analysis conducted during the interim phase has confirmed and deepened this finding. Process mapping reveals a workflow characterized by high manual effort, format variability, and the absence of feedback mechanisms. Fishbone analysis identifies the technology dimension as the most tractable point of intervention. Five Why analysis confirms that the root cause is the absence of information architecture rather than individual failure. Bottleneck analysis locates the primary constraint at the interpretation and identification stage, which is precisely the stage most amenable to AI-assisted automation. Operational gap analysis quantifies the distance between the current and desired state across six dimensions.

The ProcureIQ system concept, developed in Chapter 4, is a direct and calibrated response to these analytical findings. It is designed to address the procurement memory gap by automating the information extraction and storage steps that currently result in knowledge loss, creating a searchable repository that grows more valuable with each procurement cycle. The system is positioned as a procurement information management tool, not an autonomous procurement engine, and its scope is deliberately constrained to ensure realistic implementation within the internship timeline.

The project is well positioned for the next phase, which will involve prototype development, extraction testing, and the preparation of the final project report and IC case study. The analytical foundation is solid, the technical pathway is clear, and the operational evidence is compelling. The project is on track to produce a credible, defensible, and practically valuable contribution to both the academic and organizational contexts in which it is situated.

REFERENCES

Chopra, S., & Meindl, P. (2016). Supply chain management: Strategy, planning, and operation (6th ed.). Pearson Education.

Cousins, P., Lamming, R., Lawson, B., & Squire, B. (2008). Strategic supply management: Principles, theories and practice. Pearson Education.

Davenport, T. H., & Prusak, L. (1998). Working knowledge: How organizations manage what they know. Harvard Business School Press.

Davenport, T. H., & Ronanki, R. (2018). Artificial intelligence for the real world. Harvard Business Review, 96(1), 108–116.

European Parliament. (2016). Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data (General Data Protection Regulation). Official Journal of the European Union, L 119, 1–88.

Goldratt, E. M., & Cox, J. (2004). The goal: A process of ongoing improvement (3rd ed.). North River Press.

Harrington, H. J. (1991). Business process improvement: The breakthrough strategy for total quality, productivity, and competitiveness. McGraw-Hill.

Kusiak, A. (2018). Smart manufacturing must embrace big data. Nature, 544(7648), 23–25. https://doi.org/10.1038/544023a

Lysons, K., & Farrington, B. (2016). Procurement and supply chain management (9th ed.). Pearson Education.

Nonaka, I., & Takeuchi, H. (1995). The knowledge-creating company: How Japanese companies create the dynamics of innovation. Oxford University Press.

Ohno, T. (1988). Toyota production system: Beyond large-scale production. Productivity Press.

Radovilsky, Z., Hegde, V., & Acharya, A. (2018). Skills and competencies required for jobs of supply chain management professionals. Journal of Supply Chain and Operations Management, 16(1), 17–35.

Slack, N., Brandon-Jones, A., & Johnston, R. (2019). Operations management (9th ed.). Pearson Education.

Tadelis, S. (2012). Public procurement design: Lessons from the private sector. International Journal of Industrial Organization, 30(3), 297–302. https://doi.org/10.1016/j.ijindorg.2011.09.005

Wamba, S. F., Akter, S., Edwards, A., Chopin, G., & Gnanzou, D. (2015). How 'big data' can make big impact: Findings from a systematic review and a longitudinal case study. International Journal of Production Economics, 165, 234–246. https://doi.org/10.1016/j.ijpe.2014.12.031

Williamson, O. E. (1985). The economic institutions of capitalism: Firms, markets, relational contracting. Free Press.

ANNEXURE A: INTERIM EVIDENCE LOG

Prepared in accordance with Annexure IV of the FIIB Corporate Internship Program (AS2) requirements. The evidence log documents all artefacts generated, gathered, or planned during the interim phase of the project.

Item	Evidence Artifact	Date	What This Item Is Expected to Show
1	Current-State RFQ Workflow Process Map	May 2026	Progress snapshot documenting full procurement cycle from RFQ receipt to supplier quotation.
2	RFQ Sample Set, Stream 1 (Alibaba/Industrial)	May 2026	Data extract showing format fragmentation in platform-based industrial RFQs.
3	RFQ Sample Set, Stream 2 (Turbine/OEM Spare Parts)	May 2026	Data extract showing technical complexity and multi-reference structure of specialized RFQs.
4	Fishbone Root Cause Analysis Diagram	May–Jun 2026	Observation log identifying root causes of manual procurement information dependency.
5	5 Why Analysis, Procurement Memory Gap	May–Jun 2026	Framework application identifying systemic causes of repeated sourcing effort.
6	ProcureIQ Conceptual Architecture Diagram	Jun 2026	Draft AI system workflow showing RFQ input, extraction logic, database layer, and retrieval.
7	Procurement Information Gap Analysis Table	Jun 2026	Interim analytical table mapping identified inefficiencies to proposed system capabilities.
8	Supervisor Discussion Note, AI Concept Validation	Jun 2026	Mentor feedback note validating operational problem relevance and AI direction.

ANNEXURE B: RFQ FORMAT CLASSIFICATION MATRIX

The following matrix documents the classification of RFQ input formats encountered during the internship, their frequency of occurrence across the two procurement streams, and their implications for the ProcureIQ extraction logic.

RFQ Format	Stream 1 Frequency	Stream 2 Frequency	AI Extraction Approach	Extraction Complexity
Product photograph (label/packaging)	High	Low	GPT-4 Vision OCR + product identification	Medium
Screenshot of WhatsApp/email	High	Medium	GPT-4 Vision OCR + field mapping	Medium

PDF document (technical spec)	Low	High	PDF text extraction + GPT-4 NLP parsing	High
Excel spreadsheet (part list)	Medium	High	Structured data parsing + field mapping	Low to Medium
Plain text description	Medium	Medium	GPT-4 text completion + entity extraction	Low
Email attachment (multi-part RFQ)	Low	High	PDF/text extraction + multi-item parsing	High

The classification matrix confirms that no single extraction approach covers all RFQ formats encountered at Negbuy. The ProcureIQ system must incorporate both vision-based (image and screenshot) and text-based extraction pathways, with the appropriate pathway selected automatically based on the format of the uploaded input. This multi-modal extraction requirement is a design constraint that has been incorporated into the technical architecture described in Chapter 4.

ANNEXURE C: ANNEXURE IV, INTERIM REPORT TEMPLATE

Submitted in accordance with FIIB Corporate Internship Program requirements. Annexure IV is the official mid-course progress document.

Student Name	Vinayak Dubey
Roll Number	25PGDM170
Internship Organization	Negbuy
Corporate Mentor Name and Designation	Mr. Sayeem Salmani, Supply Chain Manager
Faculty Mentor Name	Dr. Gunjan Sood
Functional Area of Work	Procurement and Supply Chain Management, RFQ handling, supplier sourcing, quotation comparison, procurement coordination, and supplier follow-up.
Progress Summary	Workflow mapping across both procurement streams completed. Seven procurement inefficiencies identified. Fishbone, 5 Why, bottleneck, gap, and information flow analyses completed. ProcureIQ database schema and system architecture finalized. Prompt engineering exploration underway. Prototype development planned for subsequent phase.
Key Analytical Developments	Process mapping reveals seven-stage manual workflow with four delay points. Fishbone analysis confirms technology as primary

	intervention dimension. 5 Why analysis identifies absence of information architecture as root cause. Bottleneck analysis confirms RFQ interpretation as throughput constraint. Operational gap analysis quantifies six-dimension gap between current and desired state.
IC Progress Note	ProcureIQ system concept fully designed. Database schema completed (16 fields). Technical architecture confirmed (Streamlit + OpenAI API + Google Sheets). Prompt engineering exploration initiated with live RFQ samples. Prototype development to begin in subsequent phase. IC case study draft planned for Weeks 13–14.
Risks and Constraints	Limited historical data access; API cost constraints; confidentiality obligations; prototype timeline pressure. All risks mitigated through scope calibration and anonymization protocols.